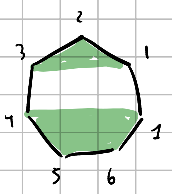


1 BP   green regions must contain 3 chords

---

1 BP   always looks like $\frac{1}{X_{14}} + \frac{1}{X_{3n}} \to 0$ on constraints

---

2 BP

$n=6$	$n=7$	$n=8$
14 15, 15 35, 3n 35	14 15, 15 35, 3n 35	14 15, 15 35, 3n 35
14 24, 24 2n, 2n 3n	14 16, 16 36, 3n 36	14 16, 16 36, 3n 36
	14 24, 24 2n, 2n 3n	14 17, 17 37, 3n 37
		14 24, 24 2n, 2n 3n

in general: $(14\ 1j,\ 1j\ 3j,\ 3j\ 3n)$ $j = 5, \dots, n-1$

$(14\ 24,\ 24\ 2n,\ 2n\ 3n)$

distinct D-subsets

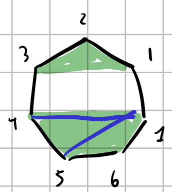
$$\frac{1}{X_{14}X_{24}} + \frac{1}{X_{24}X_{2n}} + \frac{1}{X_{2n}X_{3n}} \qquad \text{(A)}$$

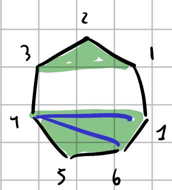
$$\left(\frac{1}{X_{14}X_{1j}} + \frac{1}{X_{3j}X_{1j}} + \frac{1}{X_{3n}X_{3j}}\right) \qquad \text{(B)}$$

$$\begin{cases} X_{3j} = X_{1j} - X_{14} & j = 5, \dots, n-1 \\ X_{2j} = X_{3j} + X_{24} & j = 5, \dots, n-1 \end{cases}$$

$$\begin{cases} X_{3n} = -X_{14} \\ X_{2n} = X_{24} - X_{14} \end{cases}$$

if X_{13} not a chord, then must have





A

Common denominator: $X_{14}X_{24}X_{2n}X_{3n}$ $X_{3n} = -X_{14}$, $X_{2n} = X_{24} - X_{14}$

$$\propto X_{2n}X_{3n} + X_{14}X_{3n} + X_{14}X_{24} = -X_{14}(\cancel{X_{24}} - \cancel{X_{14}}) + -X_{14}^2 + X_{14}\cancel{X_{24}} = 0$$

✓

(B) Common denominator: $X_{14}X_{3n}X_{1j}X_{3j}$

$$\to \alpha X_{3n}X_{3j} + X_{14}X_{3n} + X_{14}X_{1j} = -X_{14}(\cancel{X_{1j}} - \cancel{X_{14}}) - \cancel{X_{14}^2} + \cancel{X_{14}X_{1j}} = 0$$

---

3 BP

$n=6$ 14 24 15, 15 25 24, 15 25 35, 24 25 2n, 2n 25 35, 2n 3n 35

$n=7$
14 15 16, 15 16 35, 16 35 36, 35 36 3n
14 15 24, 15 24 25, 15 25 35, 24 25 2n, 25 2n 35, 2n 35 3n
14 16 24, 16 24 26, 16 26 36, 24 26 2n, 26 2n 36, 2n 36 3n

$n=8$
14 15 16, 15 16 35, 16 35 36, 35 36 3n
14 15 17, 15 17 35, 17 35 37, 35 37 3n
14 16 17, 16 17 36, 17 36 37, 36 37 3n
14 15 24, 15 24 25, 15 25 35, 24 25 2n, 25 2n 35, 2n 35 3n
14 16 24, 16 24 26, 16 25 36, 24 26 2n, 26 2n 36, 2n 36 3n
~ same with 6 → 7

$n=9$
14 15 16 15 16 35, 16 35 36, 35 36 3n
14 15 17, 15 17 35, 17 35 37, 35 37 3n
14 15 18, 15 18 35, 18 35 38, 35 38 3n
14 16 17, 16 17 36, 17 36 37, 36 37 3n
14 16 18, 15 18 36, 18 36 38, 36 38 3n
14 17 18, 15 18 37, 18 37 38, 37 38 3n

14 $1j_1$ $1j_2$ --- $\begin{cases} j_1 < j_2 \\ j_1, j_2 \in \{5, \dots, n-1\} \end{cases}$

14 15 24, 15 24 25, 15 25 35, 24 25 2n, 25 2n 35, 2n 35 3n
same with 6, then 7, then 8

only $n > 6$

pattern: (14 $1j_1$ $1j_2$, $1j_1$ $1j_2$ $3j_1$, $1j_2$ $3j_1$ $3j_2$, $3j_1$ $3j_2$ 3n) $j_1 < j_2$ $j, j_1, j_2 \in \{5, \dots, n-1\}$

(14 $1j$ 24, $1j$ 24 $2j$, $1j$ $2j$ $3j$, 24 $2j$ 2n, $2j$ 2n $3j$, 2n $3j$ 3n)

1BP • (14, 3n)

2BP • (14 $1j$, $1j$ $3j$, $3j$ 3n) $j = 5, \dots, n-1$
• (14 24, 24 2n, 2n 3n)

---

3 BP • (14 $1j_1$ $1j_2$, $1j_1$ $1j_2$ $3j_1$, $1j_2$ $3j_1$ $3j_2$, $3j_1$ $3j_2$ 3n) $j_1 < j_2$ $j_1, j_2, j \in \{5, \dots, n-1\}$
• (14 $1j$ 24, $1j$ 24 $2j$, $1j$ $2j$ $3j$, 24 $2j$ 2n, $2j$ 2n $3j$, 2n $3j$ 3n)

4BP

- (14 1j 1k 1l, 1j 1k 1l 3j, 1k 1l 3j 3k, 1l 3j 3k 3l, 3j 3k 3l 3n) $\quad j<k<l$
- (1j 1l 14 24, 1j 1l 24 2j, 1j 1l 2j 3j, 1l 24 2j 2l, 1l 2j 2l 3j, 1l 2l 3j 3l, 24 2j 2l 2n, 2j 2l 2n 3j, 2l 2n 3j 3l, 2n 3j 3l 3n) $\quad j<l$

4BP?

n = 7

```
{{{1, 4}, {1, 5}, {1, 6}, {2, 4}}, {{1, 5}, {1, 6}, {2, 4}, {2, 5}}, {{1, 5}, {1, 6}, {2, 5}, {3, 5}}, {{1, 6}, {2, 4}, {2, 5}, {2, 6}}, {{1, 6}, {2, 5}, {2, 6}, {3, 5}},
 {{1, 6}, {2, 6}, {3, 5}, {3, 6}}, {{2, 4}, {2, 5}, {2, 6}, {2, 7}}, {{2, 5}, {2, 6}, {2, 7}, {3, 5}}, {{2, 6}, {2, 7}, {3, 5}, {3, 6}}, {{2, 7}, {3, 5}, {3, 6}, {3, 7}}}
```

n = 8

```
{{{1, 4}, {1, 5}, {1, 6}, {2, 4}}, {{1, 5}, {1, 6}, {2, 4}, {2, 5}}, {{1, 5}, {1, 6}, {2, 5}, {3, 5}}, {{1, 6}, {2, 4}, {2, 5}, {2, 6}}, {{1, 6}, {2, 5}, {2, 6}, {3, 5}},
 {{1, 6}, {2, 6}, {3, 5}, {3, 6}}, {{2, 4}, {2, 5}, {2, 6}, {2, 8}}, {{2, 5}, {2, 6}, {2, 8}, {3, 5}}, {{2, 6}, {2, 8}, {3, 5}, {3, 6}}, {{2, 8}, {3, 5}, {3, 6}, {3, 8}}}
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{{{1, 4}, {1, 5}, {1, 7}, {2, 4}}, {{1, 5}, {1, 7}, {2, 4}, {2, 5}}, {{1, 5}, {1, 7}, {2, 5}, {3, 5}}, {{1, 7}, {2, 4}, {2, 5}, {2, 7}}, {{1, 7}, {2, 5}, {2, 7}, {3, 5}},
 {{1, 7}, {2, 7}, {3, 5}, {3, 7}}, {{2, 4}, {2, 5}, {2, 7}, {2, 8}}, {{2, 5}, {2, 7}, {2, 8}, {3, 5}}, {{2, 7}, {2, 8}, {3, 5}, {3, 7}}, {{2, 8}, {3, 5}, {3, 7}, {3, 8}}}
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{{{1, 4}, {1, 6}, {1, 7}, {2, 4}}, {{1, 6}, {1, 7}, {2, 4}, {2, 6}}, {{1, 6}, {1, 7}, {2, 6}, {3, 6}}, {{1, 7}, {2, 4}, {2, 6}, {2, 7}}, {{1, 7}, {2, 6}, {2, 7}, {3, 6}},
 {{1, 7}, {2, 7}, {3, 6}, {3, 7}}, {{2, 4}, {2, 6}, {2, 7}, {2, 8}}, {{2, 6}, {2, 7}, {2, 8}, {3, 6}}, {{2, 7}, {2, 8}, {3, 6}, {3, 7}}, {{2, 8}, {3, 6}, {3, 7}, {3, 8}}}
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{{{1, 4}, {1, 5}, {1, 6}, {1, 7}}, {{1, 5}, {1, 6}, {1, 7}, {3, 5}}, {{1, 6}, {1, 7}, {3, 5}, {3, 6}}, {{1, 7}, {3, 5}, {3, 6}, {3, 7}}, {{3, 5}, {3, 6}, {3, 7}, {3, 8}}}
```

n = 9

```
{{{1, 4}, {1, 5}, {1, 6}, {2, 4}}, {{1, 5}, {1, 6}, {2, 4}, {2, 5}}, {{1, 5}, {1, 6}, {2, 5}, {3, 5}}, {{1, 6}, {2, 4}, {2, 5}, {2, 6}}, {{1, 6}, {2, 5}, {2, 6}, {3, 5}},
 {{1, 6}, {2, 6}, {3, 5}, {3, 6}}, {{2, 4}, {2, 5}, {2, 6}, {2, 9}}, {{2, 5}, {2, 6}, {2, 9}, {3, 5}}, {{2, 6}, {2, 9}, {3, 5}, {3, 6}}, {{2, 9}, {3, 5}, {3, 6}, {3, 9}}}
{{{1, 4}, {1, 5}, {1, 7}, {2, 4}}, {{1, 5}, {1, 7}, {2, 4}, {2, 5}}, {{1, 5}, {1, 7}, {2, 5}, {3, 5}}, {{1, 7}, {2, 4}, {2, 5}, {2, 7}}, {{1, 7}, {2, 5}, {2, 7}, {3, 5}},
 {{1, 7}, {2, 7}, {3, 5}, {3, 7}}, {{2, 4}, {2, 5}, {2, 7}, {2, 9}}, {{2, 5}, {2, 7}, {2, 9}, {3, 5}}, {{2, 7}, {2, 9}, {3, 5}, {3, 7}}, {{2, 9}, {3, 5}, {3, 7}, {3, 9}}}
{{{1, 4}, {1, 5}, {1, 8}, {2, 4}}, {{1, 5}, {1, 8}, {2, 4}, {2, 5}}, {{1, 5}, {1, 8}, {2, 5}, {3, 5}}, {{1, 8}, {2, 4}, {2, 5}, {2, 8}}, {{1, 8}, {2, 5}, {2, 8}, {3, 5}},
 {{1, 8}, {2, 8}, {3, 5}, {3, 8}}, {{2, 4}, {2, 5}, {2, 8}, {2, 9}}, {{2, 5}, {2, 8}, {2, 9}, {3, 5}}, {{2, 8}, {2, 9}, {3, 5}, {3, 8}}, {{2, 9}, {3, 5}, {3, 8}, {3, 9}}}
{{{1, 4}, {1, 6}, {1, 7}, {2, 4}}, {{1, 6}, {1, 7}, {2, 4}, {2, 6}}, {{1, 6}, {1, 7}, {2, 6}, {3, 6}}, {{1, 7}, {2, 4}, {2, 6}, {2, 7}}, {{1, 7}, {2, 6}, {2, 7}, {3, 6}},
 {{1, 7}, {2, 7}, {3, 6}, {3, 7}}, {{2, 4}, {2, 6}, {2, 7}, {2, 9}}, {{2, 6}, {2, 7}, {2, 9}, {3, 6}}, {{2, 7}, {2, 9}, {3, 6}, {3, 7}}, {{2, 9}, {3, 6}, {3, 7}, {3, 9}}}
{{{1, 4}, {1, 6}, {1, 8}, {2, 4}}, {{1, 6}, {1, 8}, {2, 4}, {2, 6}}, {{1, 6}, {1, 8}, {2, 6}, {3, 6}}, {{1, 8}, {2, 4}, {2, 6}, {2, 8}}, {{1, 8}, {2, 6}, {2, 8}, {3, 6}},
 {{1, 8}, {2, 8}, {3, 6}, {3, 8}}, {{2, 4}, {2, 6}, {2, 8}, {2, 9}}, {{2, 6}, {2, 8}, {2, 9}, {3, 6}}, {{2, 8}, {2, 9}, {3, 6}, {3, 8}}, {{2, 9}, {3, 6}, {3, 8}, {3, 9}}}
{{{1, 4}, {1, 7}, {1, 8}, {2, 4}}, {{1, 7}, {1, 8}, {2, 4}, {2, 7}}, {{1, 7}, {1, 8}, {2, 7}, {3, 7}}, {{1, 8}, {2, 4}, {2, 7}, {2, 8}}, {{1, 8}, {2, 7}, {2, 8}, {3, 7}},
 {{1, 8}, {2, 8}, {3, 7}, {3, 8}}, {{2, 4}, {2, 7}, {2, 8}, {2, 9}}, {{2, 7}, {2, 8}, {2, 9}, {3, 7}}, {{2, 8}, {2, 9}, {3, 7}, {3, 8}}, {{2, 9}, {3, 7}, {3, 8}, {3, 9}}}
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{{{1, 4}, {1, 5}, {1, 6}, {1, 7}}, {{1, 5}, {1, 6}, {1, 7}, {3, 5}}, {{1, 6}, {1, 7}, {3, 5}, {3, 6}}, {{1, 7}, {3, 5}, {3, 6}, {3, 7}}, {{3, 5}, {3, 6}, {3, 7}, {3, 9}}}
{{{1, 4}, {1, 5}, {1, 6}, {1, 8}}, {{1, 5}, {1, 6}, {1, 8}, {3, 5}}, {{1, 6}, {1, 8}, {3, 5}, {3, 6}}, {{1, 8}, {3, 5}, {3, 6}, {3, 8}}, {{3, 5}, {3, 6}, {3, 8}, {3, 9}}}
{{{1, 4}, {1, 5}, {1, 7}, {1, 8}}, {{1, 5}, {1, 7}, {1, 8}, {3, 5}}, {{1, 7}, {1, 8}, {3, 5}, {3, 7}}, {{1, 8}, {3, 5}, {3, 7}, {3, 8}}, {{3, 5}, {3, 7}, {3, 8}, {3, 9}}}
{{{1, 4}, {1, 6}, {1, 7}, {1, 8}}, {{1, 6}, {1, 7}, {1, 8}, {3, 6}}, {{1, 7}, {1, 8}, {3, 6}, {3, 7}}, {{1, 8}, {3, 6}, {3, 7}, {3, 8}}, {{3, 6}, {3, 7}, {3, 8}, {3, 9}}}
```

(1j 1l 14 24, 1j 1l 24 2j, 1j 1l 2j 3j, 1l 24 2j 2l, 1l 2j 2l 3j, 1l 2l 3j 3l, 24 2j 2l 2n, 2j 2l 2n 3j, 2l 2n 3j 3l, 2n 3j 3l 3n) $\quad j<l$

(14 1j 1k 1l, 1j 1k 1l 3j, 1k 1l 3j 3k, 1l 3j 3k 3l, 3j 3k 3l 3n) $\quad j<k<l$

next goal: horizontal pattern

(14 1j, 1j 3j, 3j 3n)

(14 1i 1j, 1i 1j 3i, 1j 3i 3j, 3i 3j 3n)

(14 $1j_1$ $1j_2$ $1j_3$, $1j_1$ $1j_2$ $1j_3$ $3j_1$, $1j_2$ $1j_3$ $3j_1$ $3j_2$, $1j_3$ $3j_1$ $3j_2$ $3j_3$, $3j_1$ $3j_2$ $3j_3$ 3n)

$s = 3$

$$\boxed{\frac{1}{X_{14}\prod_{a=1}^{s} X_{1j_a}} + \frac{1}{X_{3n}\prod_{a=1}^{s} X_{3j_a}} + \sum_{i=1}^{s} \frac{1}{\prod_{a=i}^{s} X_{1j_a} \prod_{b=1}^{i} X_{3j_b}}}$$

one of the patterns

if $j_0 = 4$ $\quad j_{s+1} = n$

$$\rightarrow \sum_{i=0}^{s+1} \frac{1}{\prod_{a=i}^{s} X_{1j_a} \prod_{b=1}^{i} X_{3j_b}}$$

→ verifies using same method as yesterday.

$s=2$

$(14\,24,\ 24\,2n,\ 2n\,3n)$

$(14\,1j\,24,\ 1j\,24\,2j,\ 1j\,2j\,3j,\ 24\,2j\,2n,\ 2j\,2n\,3j,\ 2n\,3j\,3n)$

$(1j_1\,1j_2\,1j_0\,2j_0,\ 1j_1\,1j_2\,2j_0\,2j_1,\ 1j_1\,1j_2\,2j_1\,3j_1,\ 1j_2\,2j_0\,2j_1\,2j_2,\ 1j_2\,2j_1\,2j_2\,3j_1,\ 1j_2\,2j_2\,3j_1\,3j_2,\ 2j_0\,2j_1\,2j_2\,2j_3,$
$2j_1\,2j_2\,2j_3\,3j_1,\ 2j_2\,2j_{s+1}\,3j_1\,3j_2,\ 2j_{s+1}\,3j_1\,3j_2\,3j_{s+1})$ $\quad j < l$

$$\prod_{a=0}^{s} X_{1j_a} \prod_{b=0}^{0} X_{2j_b} \qquad \prod_{a=1}^{s} X_{1j_a} \prod_{b=0}^{1} X_{2j_b} \qquad \prod_{a=2}^{s} X_{1j_a} \prod_{b=0}^{2} X_{2j_b} \rightarrow \boxed{\prod_{a=i}^{s} X_{1j_a} \prod_{b=0}^{i} X_{2j_b}} \quad i = 0, \dots, s$$

$$\prod_{a=1}^{s+1} X_{3j_a} \prod_{b=s+1}^{s+1} X_{2j_b} \qquad \prod_{a=1}^{2} X_{3j_a} \prod_{b=2}^{s+1} X_{2j_b} \qquad \prod_{a=1}^{1} X_{3j_a} \prod_{b=1}^{s+1} X_{2j_b} \rightarrow \boxed{\prod_{a=1}^{i} X_{3j_a} \prod_{b=i}^{s+1} X_{2j_b}} \quad i = 1, \dots, s+1$$

$$\prod_{a=1}^{2} X_{1j_a} \prod_{b=1}^{1} X_{2j_b} \prod_{c=1}^{1} X_{3j_c} \qquad \prod_{a=2}^{2} X_{1j_a} \prod_{b=1}^{2} X_{2j_b} \prod_{c=1}^{1} X_{3j_c} \qquad \prod_{a=2}^{2} X_{1j_a} \prod_{b=2}^{2} X_{2j_b} \prod_{c=1}^{2} X_{3j_c}$$

$$\rightarrow \boxed{\prod_{a=i_1}^{s} X_{1j_a} \prod_{b=i_2}^{i_1} X_{2j_b} \prod_{c=1}^{i_2} X_{3j_c}} \quad i_1, i_2 \in \{1, \dots, s\} \text{ with } i_2 \le i_1$$

$$\prod_{a=0}^{s+1} X_{2j_a}$$

$$\rightarrow \boxed{\begin{aligned} &\underset{1}{\left(\prod_{a=0}^{s+1} X_{2j_a}\right)^{-1}} + \overset{2}{\sum_{i=0}^{s} \left(\prod_{a=i}^{s} X_{1j_a} \prod_{b=0}^{i} X_{2j_b}\right)^{-1}} + \overset{3}{\sum_{i=1}^{s+1} \left(\prod_{a=1}^{i} X_{3j_a} \prod_{b=i}^{s+1} X_{2j_b}\right)^{-1}} \\ &+ \sum_{i_1=1}^{s} \sum_{i_2=1}^{i_1} \left(\prod_{a=i_1}^{s} X_{1j_a} \prod_{b=i_2}^{i_1} X_{2j_b} \prod_{c=1}^{i_2} X_{3j_c}\right)^{-1} \ 4 \end{aligned}}$$

Common denominator $\left[X_{14}\, X_{3n}\, X_{24}\, X_{2n} \prod_{e=1}^{s} \left(X_{1je}\, X_{2je}\, X_{3je} \right) \right] = D$

① $X_{14}\, X_{3n} \prod_{e=1}^{s} \left(X_{1je}\, X_{3je} \right)$

② $$\underbrace{\frac{D}{X_{14}\, X_{24} \prod_{e=1}^{s} X_{1je}}}_{i=0} + \sum_{i=1}^{s} \frac{D}{X_{24} \prod_{e=i}^{s} X_{1je} \prod_{b=1}^{i} X_{2jb}} =$$

$$= X_{3n}\, X_{2n} \prod_{e=1}^{s} \left(X_{2je}\, X_{3je} \right) + \sum_{i=1}^{s} X_{14}\, X_{3n}\, X_{2n} \prod_{e=1}^{i-1} X_{1je} \prod_{b=i+1}^{s} X_{2jb} \prod_{c=1}^{s} X_{3jc}$$

③ $$\underbrace{\frac{D}{X_{3n}\, X_{2n} \prod_{a=1}^{s} X_{3je}}}_{i=s+1} + \sum_{i=1}^{s} \frac{D}{X_{2n} \prod_{e=1}^{i} X_{3je} \prod_{b=i}^{s} X_{2jb}} =$$

$$= X_{14}\, X_{24} \prod_{e=1}^{s} \left(X_{2je}\, X_{1je} \right) + \sum_{i=1}^{s} X_{14}\, X_{3n}\, X_{24} \prod_{e=1}^{s} X_{1je} \prod_{b=1}^{i-1} X_{2jb} \prod_{c=i+1}^{s} X_{3jc}$$

④ $$= \sum_{i_1=1}^{s} \sum_{i_2=1}^{i_1} \frac{X_{14}\, X_{3n}\, X_{24}\, X_{2n} \prod_{e=1}^{s} \left(X_{1je}\, X_{2je}\, X_{3je} \right)}{\prod_{a=i_1}^{s} X_{1ja} \prod_{b=i_2}^{i_1} X_{2jb} \prod_{c=1}^{i_2} X_{3jc}} =$$

$$= X_{14}\, X_{3n}\, X_{24}\, X_{2n} \prod_{a=1}^{i_1-1} X_{1ja} \prod_{c=i_2+1}^{s} X_{3jc} \prod_{b=1}^{i_2-1} X_{2jb} \prod_{b=i_1+1}^{s} X_{2jb}$$

Controls: $X_{3n} = -X_{14}$ $\quad X_{2n} = X_{24} - X_{14}$ $\quad X_{3j} = X_{1j} - X_{14}$ $\quad X_{2j} = X_{1j} + X_{24} - X_{14}$

$$DS = X_{14}\, X_{3n} \prod_{e=1}^{s} \left(X_{1je}\, X_{3je} \right) + X_{3n}\, X_{2n} \prod_{e=1}^{s} \left(X_{2je}\, X_{3je} \right) +$$

$$\sum_{i=1}^{s} X_{14}\, X_{3n}\, X_{2n} \prod_{e=1}^{i-1} X_{1je} \prod_{b=i+1}^{s} X_{2jb} \prod_{c=1}^{s} X_{3jc} + X_{14}\, X_{24} \prod_{e=1}^{s} \left(X_{2je}\, X_{1je} \right)$$

$$+ \sum_{i=1}^{s} X_{14}\, X_{3n}\, X_{24} \prod_{e=1}^{s} X_{1je} \prod_{b=1}^{i-1} X_{2jb} \prod_{c=i+1}^{s} X_{3jc}$$

$$+ X_{14}\, X_{3n}\, X_{24}\, X_{2n} \prod_{a=1}^{i_1-1} X_{1ja} \prod_{c=i_2+1}^{s} X_{3jc} \prod_{b=1}^{i_2-1} X_{2jb} \prod_{b=i_1+1}^{s} X_{2jb}$$

impose

factor out X_{14}

$$DS \propto -X_{14}\prod_{e=1}^{s}\left(X_{1j_e}\left(X_{1j_e}-X_{14}\right)\right) - \left(X_{24}-X_{14}\right)\prod_{e=1}^{s}\left(X_{1j_e}-X_{14}+X_{c4}\right)\left(X_{1j_e}-X_{14}\right)$$

$$-\sum_{i=1}^{s} X_{14}\left(X_{24}-X_{14}\right)\prod_{e=1}^{i-1}X_{1j_e}\prod_{b=i+1}^{s}\left(X_{1j_b}-X_{14}+X_{c4}\right)\prod_{c=1}^{s}\left(X_{1j_c}-X_{14}\right)$$

$$+\; X_{24}\prod_{e=1}^{s}X_{1j_e}\left(X_{1j_e}-X_{14}+X_{c4}\right)$$

$$-\sum_{i=1}^{s} X_{14}X_{24}\prod_{e=1}^{s}X_{1j_e}\prod_{b=1}^{i-1}\left(X_{1j_b}-X_{14}+X_{c4}\right)\prod_{c=i+1}^{s}\left(X_{1j_c}-X_{14}\right)$$

$$-\sum_{i_1=1}^{s}\sum_{i_2=1}^{i_1} X_{14}X_{24}\left(X_{24}-X_{14}\right)\prod_{a=1}^{i_1-1}X_{1j_a}\prod_{c=i_2+1}^{s}\left(X_{1j_c}-X_{14}\right)\prod_{b=1}^{i_2-1}\left(X_{1j_b}-X_{14}+X_{c4}\right)\prod_{b=i_1+1}^{s}\left(X_{1j_b}-X_{14}+X_{c4}\right)$$

$$= -X_{14}X_{1j_1}\left(X_{1j_1}-X_{14}\right)\prod_{e=2}^{s}\left(X_{1j_e}\left(X_{1j_e}-X_{14}\right)\right) +$$

$$-\left(X_{24}-X_{14}\right)\left(X_{1j_1}-X_{14}\right)\left(X_{1j_1}-X_{14}+X_{24}\right)\prod_{e=2}^{s}\left(X_{1j_e}-X_{14}+X_{c4}\right)\left(X_{1j_e}-X_{14}\right)$$

$$-X_{14}\left(X_{24}-X_{14}\right)\left(X_{1j_1}-X_{14}\right)\prod_{b=2}^{s}\left(X_{1j_b}-X_{14}+X_{c4}\right)\prod_{c=2}^{s}\left(X_{1j_c}-X_{14}\right)$$

$$-\sum_{i=2}^{s} X_{14}\left(X_{24}-X_{14}\right)X_{1j_1}\left(X_{1j_1}-X_{14}\right)\prod_{e=2}^{i-1}X_{1j_e}\prod_{b=i+1}^{s}\left(X_{1j_b}-X_{14}+X_{c4}\right)\prod_{c=2}^{s}\left(X_{1j_c}-X_{14}\right)$$

$$+X_{24}X_{1j_1}\left(X_{1j_1}-X_{14}+X_{24}\right)\prod_{e=2}^{s}X_{1j_e}\left(X_{1j_e}-X_{14}+X_{c4}\right)$$

$$-X_{14}X_{24}X_{1j_1}\prod_{c=2}^{s}\left(X_{1j_c}-X_{14}\right)$$

$$-\sum_{i=2}^{s} X_{14}X_{24}X_{1j_1}\left(X_{1j_1}-X_{14}+X_{24}\right)\prod_{e=2}^{s}X_{1j_e}\prod_{b=2}^{i-1}\left(X_{1j_b}-X_{14}+X_{c4}\right)\prod_{c=i+1}^{s}\left(X_{1j_c}-X_{14}\right)$$

$(i_1=1,\ i_2=1)$

$$-X_{14}X_{24}\left(X_{24}-X_{14}\right)\prod_{c=2}^{s}\left(X_{1j_c}-X_{14}\right)\prod_{b=2}^{s}\left(X_{1j_b}-X_{14}+X_{c4}\right)$$

$\sum_{i_1>1}$ $i_2=1$

$$-\sum_{i_1=2}^{s} X_{14}X_{24}\left(X_{24}-X_{14}\right)X_{1j_1}\prod_{a=2}^{i_1-1}X_{1j_a}\prod_{c=2}^{s}\left(X_{1j_c}-X_{14}\right)\prod_{b=2}^{s}\left(X_{1j_b}-X_{14}+X_{c4}\right)$$

$$- X_{1j_1} \left(X_{1j_1} - X_{14} + X_{24}\right) \sum_{i_1=2}^{s} \sum_{i_2=2}^{i_1} X_{14} X_{24} \left(X_{24} - X_{14}\right) \prod_{a=2}^{i_1-1} X_{1j_a} \prod_{c=i_2+1}^{s} \left(X_{1j_c} - X_{14}\right)$$

$$* \prod_{b=2}^{i_2-1} \left(X_{1j_b} - X_{14} + X_{24}\right) \prod_{b=i_1+1}^{s} \left(X_{1j_b} - X_{14} + X_{24}\right)$$

$$-X_{14} \prod_{e=1}^{s} \left(X_{1j_e} \left(X_{1j_e} - X_{14}\right)\right) \quad + X_{24} X_{1j_1} \left(X_{1j} - X_{14} + X_{24}\right) \prod_{e=2}^{s} X_{1j_e} \left(X_{1j_e} - X_{14} + X_{24}\right)$$

$$- X_{14} X_{24} X_{1j_1} \prod_{c=2}^{s} \left(X_{1j_c} - X_{14}\right)$$